Red Hat

RH124

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Course Outlines

Day	Content
Day1	Get Started with Red Hat Enterprise Linux Access the Command Line Manage Files from the Command Line
Day2	Get Help in Red Hat Enterprise Linux Create, View, and Edit Text Files Manage Local Users and Groups
Day3	Control Access to Files Manage Linux Processes Monitor Control Services and Daemons
Day4	Configure and Secure SSH Analyzing and Storing Logs Manage Networking
Day5	Archiving and Transferring Files Install and Update Software Packages Access Linux File Systems

What about Today?

- Get Started with Red Hat Enterprise Linux
- Access the Command Line
- Manage Files from the Command Line

Get Started with Red Hat Enterprise Linux

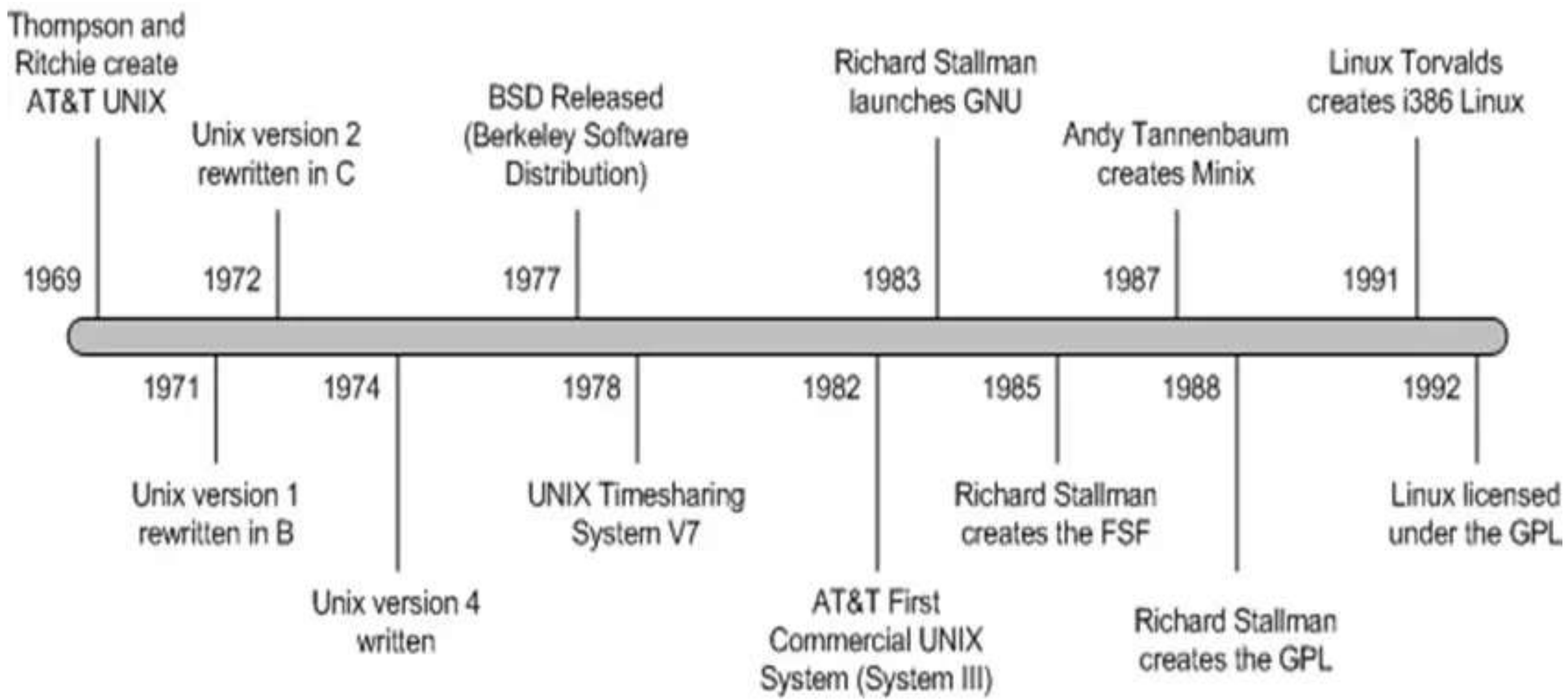
Goal

Define open source, Linux, Linux distributions, and Red Hat Enterprise Linux

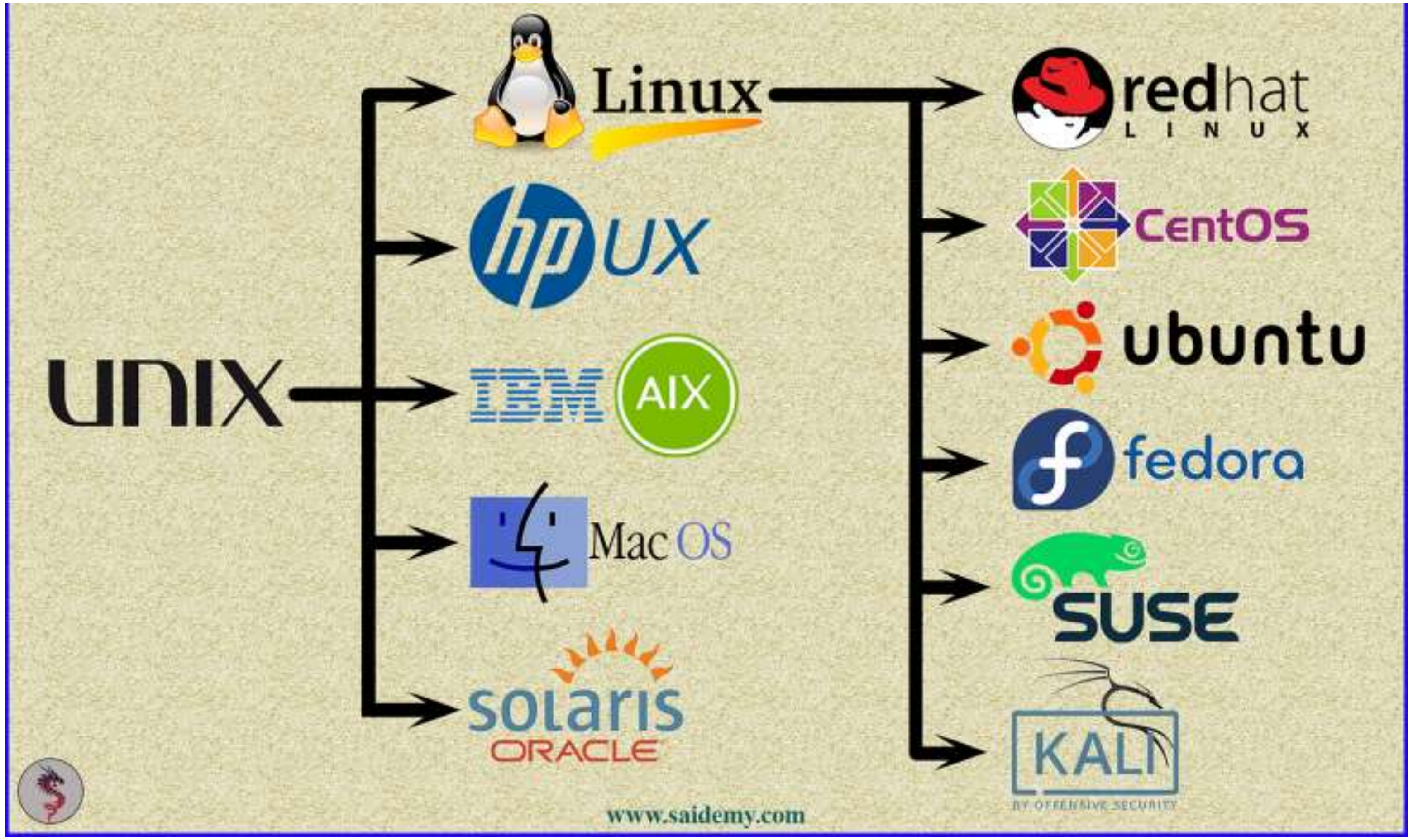
Objectives

- Explain the purpose of open source, Linux, Linux distributions, and Red Hat Enterprise Linux

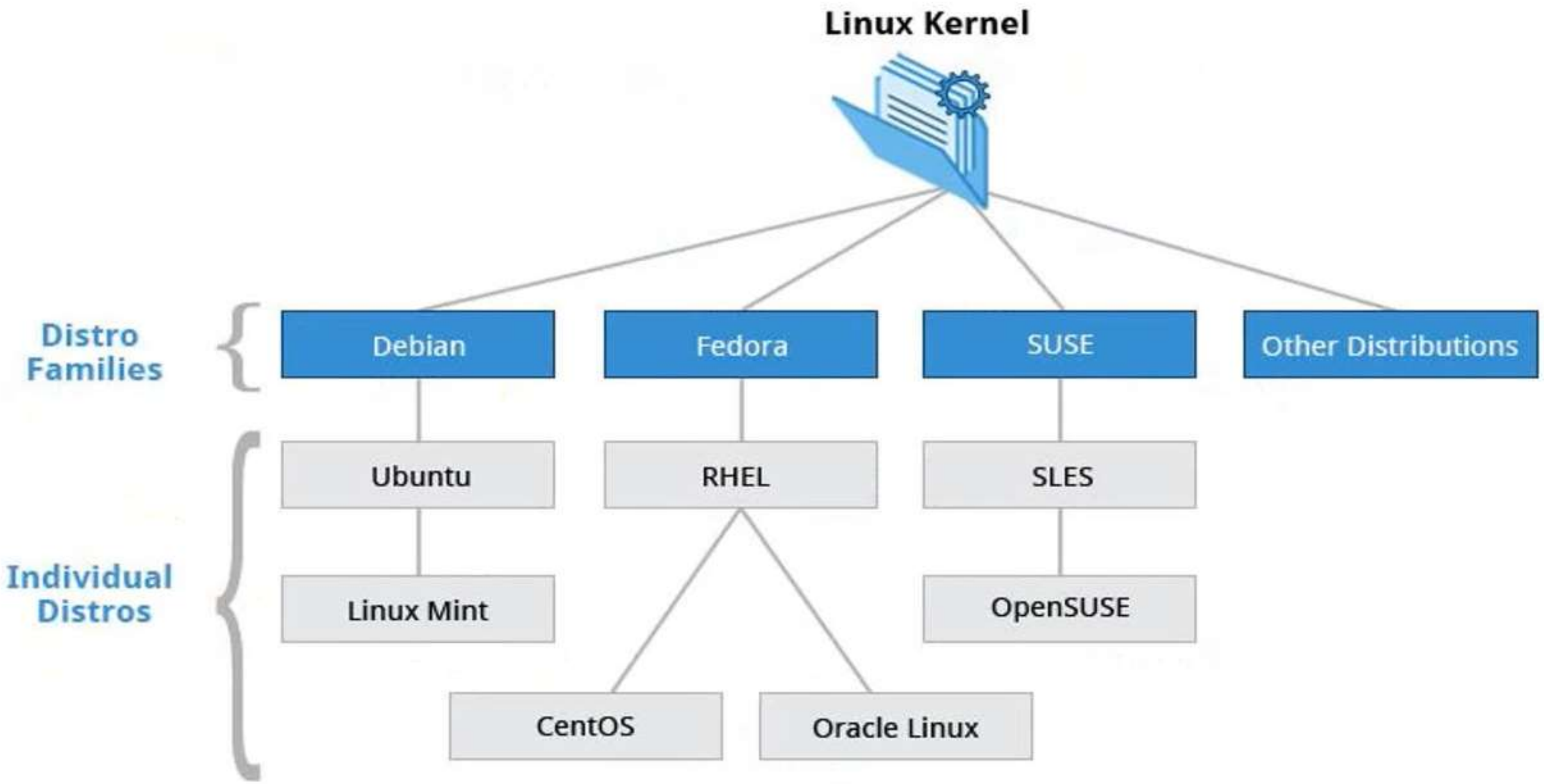
LINUX History



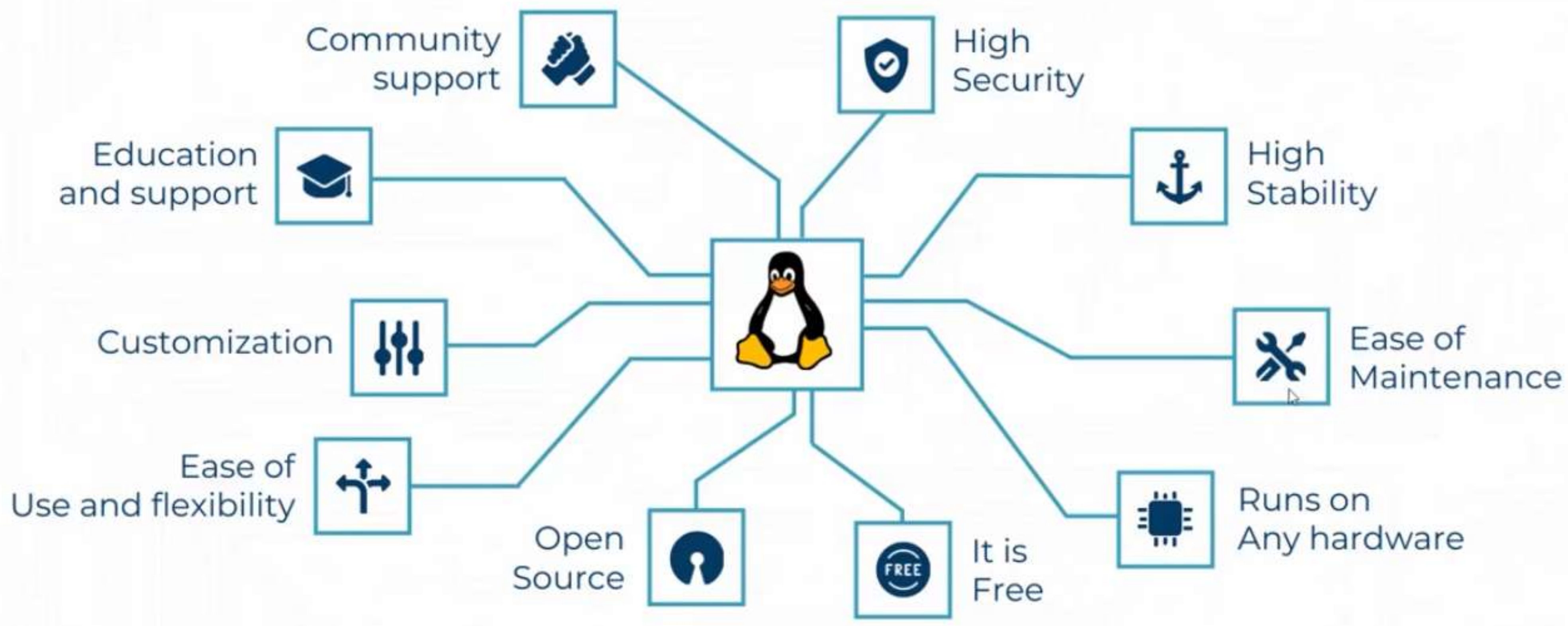
Cont. LINUX Distributions



Cont. LINUX Distributions

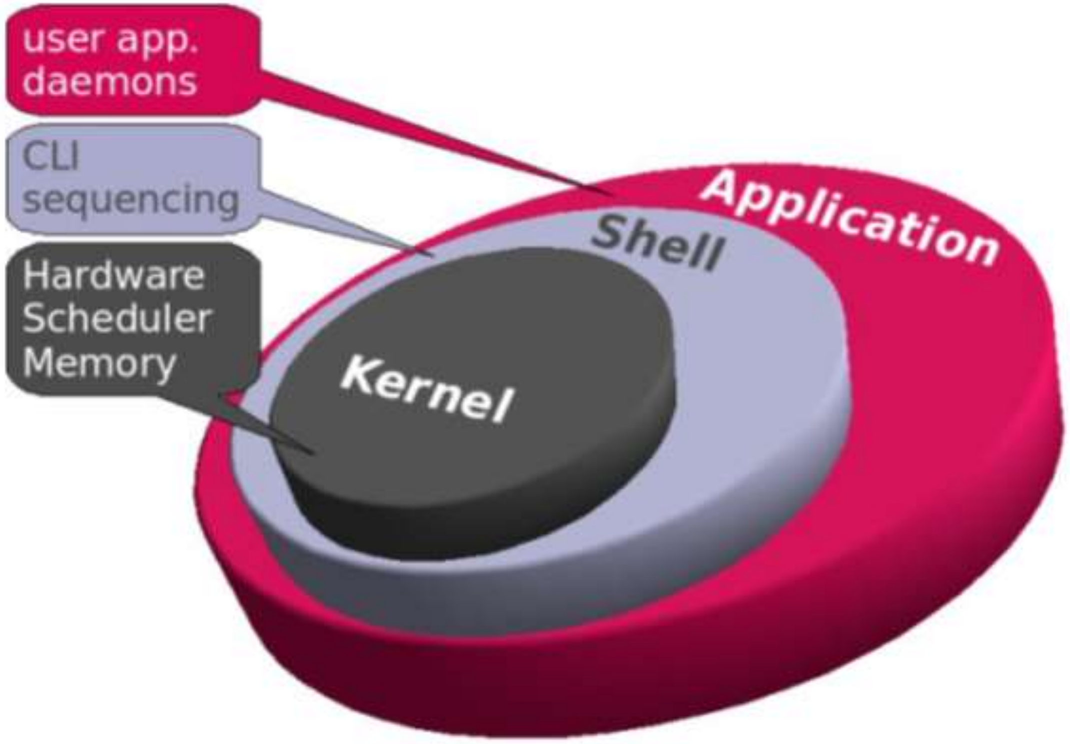


Why LINUX?


Community support
High Security
Education and support
High Stability
Customization
Ease of Maintenance
Ease of Use and flexibility
Open Source
It is Free
Runs on Any hardware
FREE

Linux Components

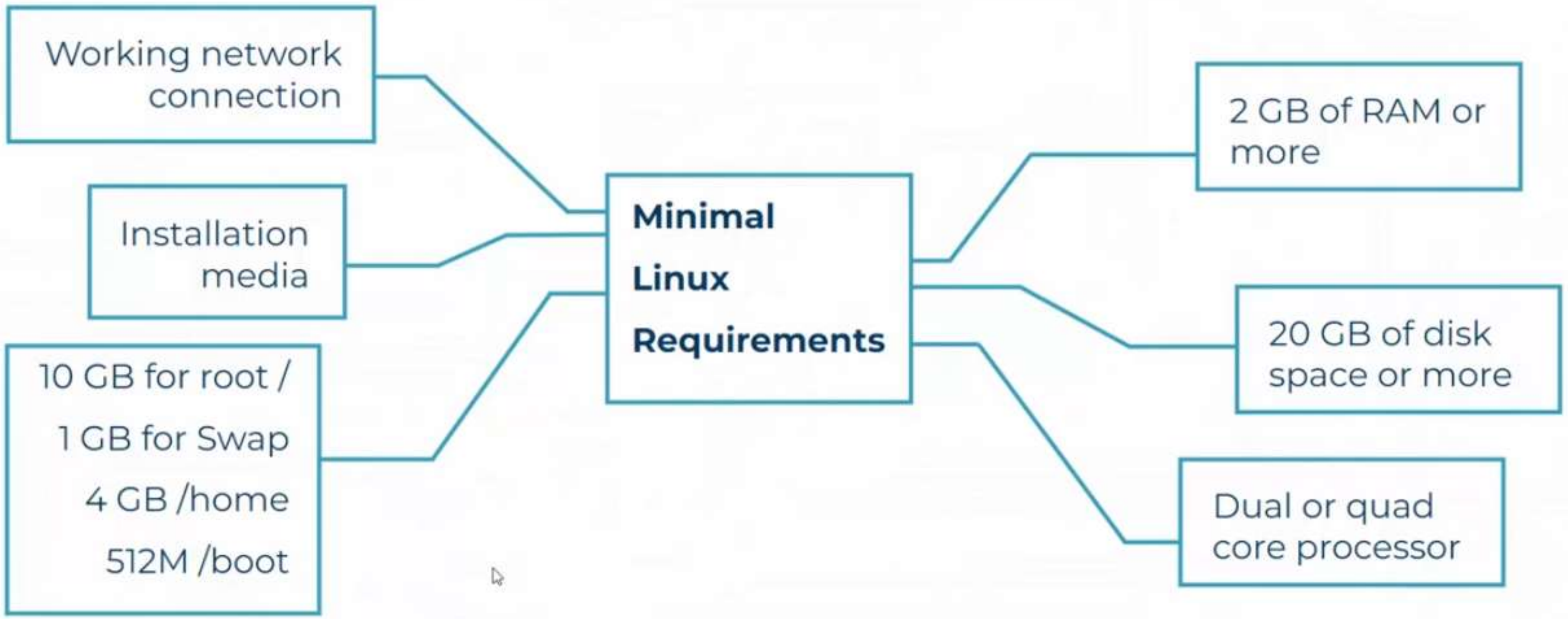
- **Kernel**
  - Is the core of the operating system.
  - Contains components like device drivers.
  - It loads into RAM when the machine boots and stays resident in RAM until the machine powers off.



Linux Components

- **Shell**
  - C shell , ksh , Bash
  - Provides an interface by which the user can communicate with
  - the kernel.
  - “bash” is the most commonly used shell on Linux.
  - The shell parses commands entered by the user and translates them into logical segments to be executed by the kernel or other utilities.

Minimum Requirements for RHEL 9



Hands-on working …

Access the Command Line

Goal

Log in to a Linux system and run simple commands from the shell

Objectives

- Log in to a Linux system and run simple commands from the shell
- Log in to the Linux system with the GNOME desktop environment to run commands from a shell prompt in a terminal program
- Save time when running commands from a shell prompt with Bash shortcuts

FROM GRAPHICAL USER INTERFACE (GUI).

REMOTE ACCESS OVER SECURE SHELL (SSH).

Access the command line

Running Commands

- Commands have the following syntax:
    - command [options] [arguments]
- Each item is separated by a space.
- Options modify the command's behavior.
- Arguments are file names or other information needed by the command.
- Separate commands with a semicolon (;).
- Example: usermod –L omar

Right

- $ ls -l /dev
- $ ls -a /dev
- $ mail -s test root
- $ who -u
- $ ls -l -d
- $ ls -ld

Wrong

- $ ls - l /dev
- $ ls-a /dev
- $ mail test root -s
- $ -u who
- $ ls -l-d
- $ ls -l d


Hands-on working
…

Shell Shortcuts

Ctrl+A	Jump to the beginning of the command line.
Ctrl+E	Jump to the end of the command line.
Ctrl+U	Clear from the cursor to the beginning of the command line.
Ctrl+K	Clear from the cursor to the end of the command line.
Ctrl+LeftArrow	Jump to the beginning of the previous word on the command line.
Ctrl+RightArrow	Jump to the end of the next word on the command line.
Ctrl+R	Search the history list of commands for a pattern.

Manage Files from the Command Line

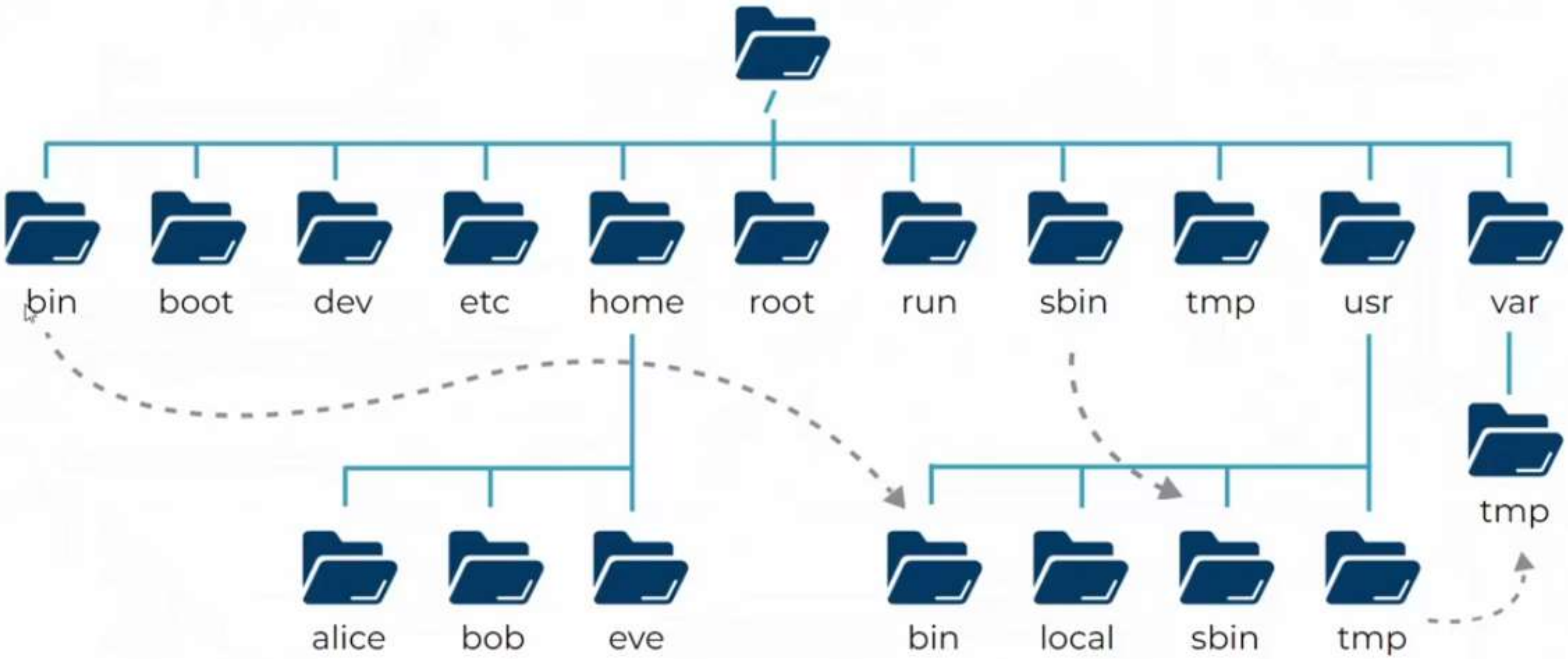
Goal

Copy, move, create, delete, and organize files from the Bash shell

Objectives

- Describe how Linux organizes files, and the purposes of various directories in the file-system hierarchy.
- Specify the absolute location and relative location of files to the current working directory, determine and change the working directory, and list the contents of directories.
- Create, copy, move, and remove files and directories.
- Create multiple file name references to the same file with hard links and symbolic (or "soft") links.
- Efficiently run commands that affect many files by using pattern-matching features of the Bash shell.

File System



bin
boot
dev
etc
home
root
run
sbin
tmp
usr
var
tmp
alice
bob
eve
bin
local
sbin
tmp

Important Directories

LOCATION	PURPOSE
/usr	Installed software, shared libraries, include files, and read-only program data. Important subdirectories include: • **/usr/bin**: User commands. • **/usr/sbin**: System administration commands. • **/usr/local**: Locally customized software.
/etc	Configuration files specific to this system.
/var	Variable data specific to this system that should persist between boots. Files that dynamically change, such as databases, cache directories, log files, printer-spooled documents, and website content may be found under **/var**.
/run	Runtime data for processes started since the last boot. This includes process ID files and lock files, among other things. The contents of this directory are recreated on reboot. This directory consolidates **/var/run** and **/var/lock** from earlier versions of Red Hat Enterprise Linux.

Important Directories

/home	*Home directories* are where regular users store their personal data and configuration files.
/root	Home directory for the administrative superuser, `root`.
/tmp	A world-writable space for temporary files. Files which have not been accessed, changed, or modified for 10 days are deleted from this directory automatically. Another temporary directory exists, **/var/tmp**, in which files that have not been accessed, changed, or modified in more than 30 days are deleted automatically.
/boot	Files needed in order to start the boot process.
/dev	Contains special *device files* that are used by the system to access hardware.

File Types

Symbol	Meaning
-	Regular file
d	Directory
l	Link
c	Special File
s	Socket
p	Named Pipe
b	Block Device

Rules for naming files

- **Should**
  - Be descriptive
  - Only alphanumeric characters

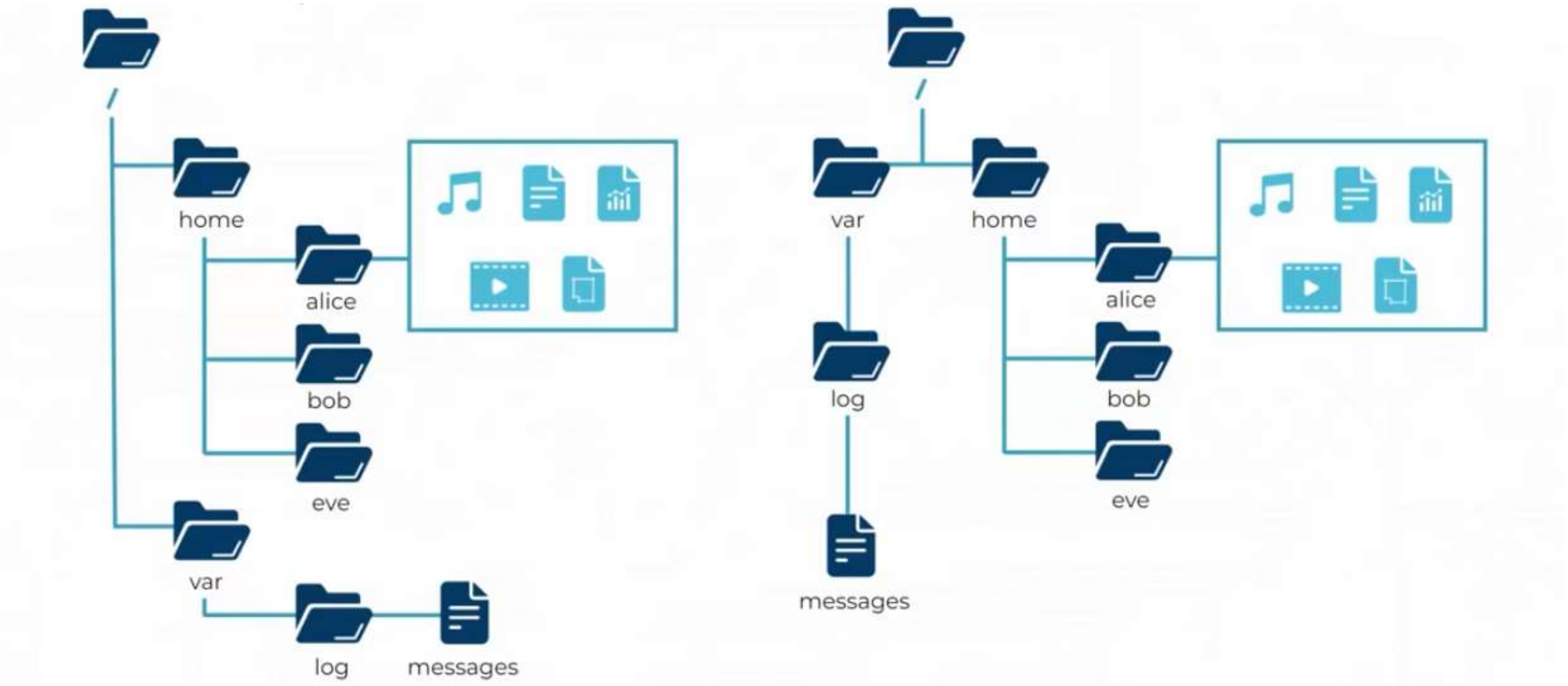
    UPERCASE, lowercase, number, @, _

- **Should Not**
  - Include embedded blanks
  - Contain shell metacharacters

    * ? > < / ; & ! [ ] | \ ' " ( ) { }

- Are case sensitive.
- Filenames starting with a "."are hidden.
- Maximum number of character for a filename is 255.

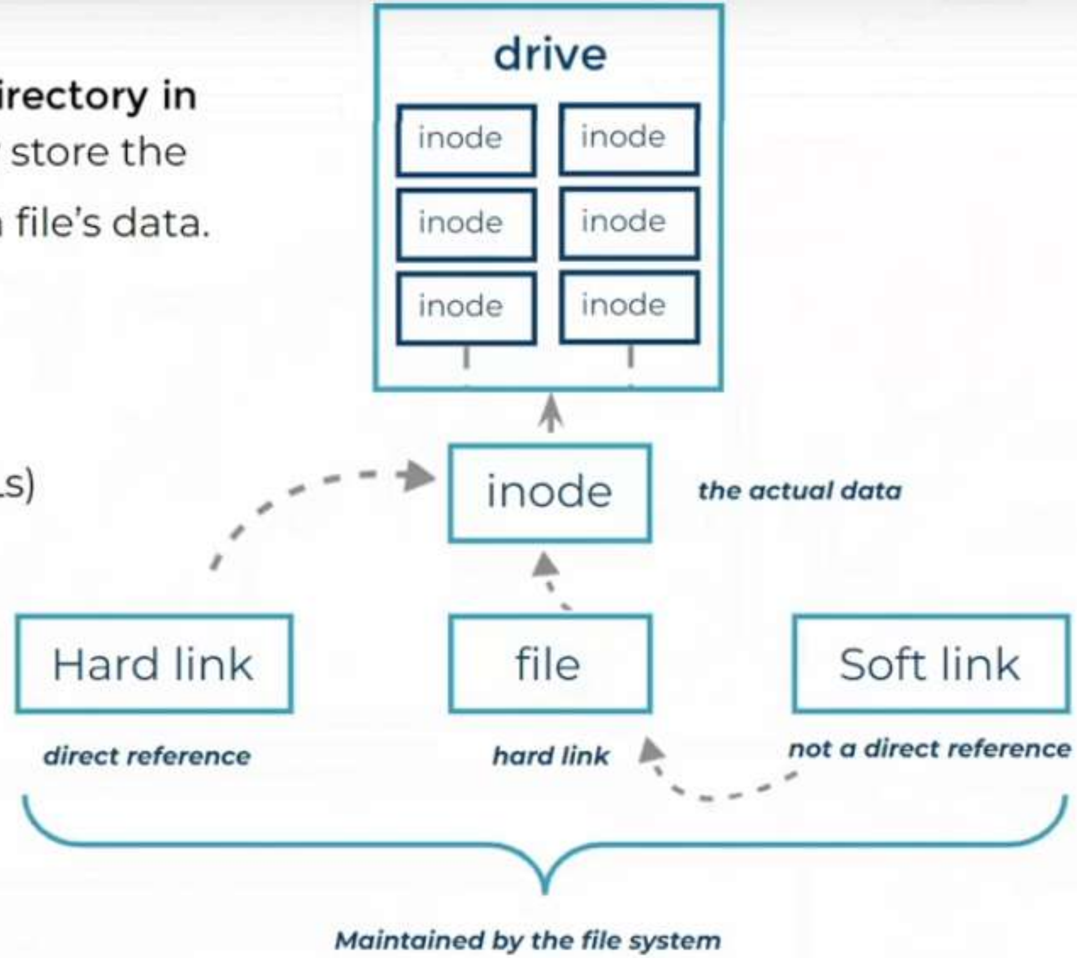
Absolute path & Relative path



What is inode on Linux Systems

Linux allocate an index node (inode) for every file and directory in filesystem. Inodes do not store actual data. Instead, they store the metadata where you can find the storage blocks of each file's data.

- The following metadata exists in an inode
  - File type
  - Permissions
  - Hard links count
  - Owner ID
  - Group ID
  - Size of file
  - Time stamp (access time)
  - Time stamp (modification time)
  - Soft/Hard Links
  - Access Control List (ACLs)



Soft Link	Hard Link
Soft link is an alias to the original file similar to the shortcut feature in Windows OS.	Hard link is the exact replica of the original file it is pointing to.
It just contain the location to the original file but not the actual data.	It contains the actual content of the file.
Soft links have different Inode values pointing to the original value.	Hard links share the same Inode value pointing to the same file location.
Links can be created across filesystem.	Links cannot be created outside the filesystem.
The link becomes inaccessible when the original file is removed.	Changes in the hard linked file will reflect in the other files.
Soft links can link both to a file or a directory.	Hard links can only link to a file, not a directory.

n matching (Regex)

PATTERN	MATCHES
*	Any string of zero or more characters.
?	Any single character.
[*abc...*]	Any one character in the enclosed class (between the square brackets).
[!*abc...*]	Any one character *not* in the enclosed class.
[^*abc...*]	Any one character *not* in the enclosed class.
[[:alpha:]]	Any alphabetic character.
[[:lower:]]	Any lowercase character.
[[:upper:]]	Any uppercase character.
[[:alnum:]]	Any alphabetic character or digit.
[[:punct:]]	Any printable character not a space or alphanumeric.
[[:digit:]]	Any single digit from 0 to 9.
[[:space:]]	Any single white space character. This may include tabs, newlines, carriage returns, form feeds, or spaces.

Regular Expressions

Symbol	Descriptions
.	replaces any character
^	matches start of string
$	matches end of string
*	matches up zero or more times the preceding character
\	Represent special characters
()	Groups regular expressions
?	Matches up exactly one character


Any questions ??

Thnak you